

## AquaDuck AI

Design to clean our rivers, monitor water quality  
Identify pollution sources & generate real time  
environmental intelligence using Edge AI.

Autonomous Navigation → (Using GPS, IMU, digital  
compass, LIDAR, ultrasonic  
sensors & cameras, it  
determines location & maps surroundings)

↓  
Jetson Orin nano processes

Sensors data & ESP32 controls motor & thrusters



AI-Based waste detection.



Waste collection.



AI based waste segregation.



Algae collection



Water quality monitoring



Edge AI water analysis



Pollution source Localization.



Environmental intelligence



Data upload & Dashboard



Autonomous Return